Myoelectric Pattern Recognition for Controlling a Robotic Hand: A Feasibility Study in Stroke

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Abstract*—*Objective: **Myoelectric pattern recognition has been successfully applied as a human-machine interface to control robotic devices such as prostheses and exoskeletons, significantly improving the dexterity of myoelectric control. This study investigates the feasibility of applying myoelectric pattern recognition for controlling a robotic hand in stroke patients.** ***Methods:*** **Myoelectric pattern recognition of six hand motion patterns was performed using forearm electromyogram signals in paretic side of eight stroke subjects. Both the random cross validation (RCV) and the chronological handout validation (CHV) were applied to assess the offline myoelectric pattern recognition performance. Experiments on real-time myoelectric pattern recognition control of an exoskeleton robotic hand were also performed.** ***Results:*** **An average classification accuracy of 84.1% (the mean value from two different classifiers) and individual subject differences were observed in the offline myoelectric pattern recognition analysis using the RCV, while the accuracy decreased to 65.7% when the CHV was used. The stroke subjects achieved an average accuracy of 61.3 $\pm$ 20.9% for controlling the robotic hand. However, our study did not reveal a clear correlation between the real-time control accuracy and the offline myoelectric pattern recognition performance, or any specific characteristics of the stroke subjects.** ***Conclusion:*** **The findings suggest that it is feasible to apply myoelectric pattern recognition to control the robotic hand in some but not all of the stroke patients. Each stroke subject should be individually online tested for the feasibility of applying myoelectric pattern recognition control for robot-assisted rehabilitation.**

***Index Terms*—EMG, myoelectric pattern recognition, real-time control, hand exoskeleton, stroke rehabilitation.**

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I. Introduction

HAND dysfunction (e.g., paresis, paralysis, and spasticity) is one of common motor symptoms among chronic stroke patients, affecting their activity of daily living and quality of life. Many studies have been performed to improve motor function of stroke patients, mainly in two categories: stimulation-based [1]–[4] and robot-based [5]–[8] approaches. A typical stimulation-based approach delivers electrical stimulations to activate one or more muscles so that a patient can perform desired movements better [3]. A typical robot-based approach usually applies an exoskeleton to assist a patient to perform desired movements [8]. Both approaches can be used to provide motor recovery and motor learning in activities of daily living, as well as repetitive training for functional rehabilitation [3], [7], [8]. Although these approaches are powerful and effective, a control system is required in order to provide a patient the desired assistance at the right time. An accurate and flexible intent or movement detection system is therefore critical, since its ability to distinguish multiple movements to a large extent determines the quality of the training or treatments. Electromyogram (EMG) is widely used to trigger or control a robotic device [2], [3], [9]–[11] because the user's motion intents can be extracted from EMG signals. However, existing studies mainly applied a conventional on-off [8], [9] or proportional [10], [12] control strategy, which can control only one or two degrees of freedom (DOF) at a time. Such a control strategy is often used to control proximal or large joint movements (such as the elbow and the wrist) that have fewer DOFs [5], [13]–[15]. Considering that a human hand has up to 27 DOFs [16], it is a challenge to regain hand function by using the conventional control strategy.

Myoelectric pattern recognition, based on the assumption that motion intents or motion patterns can be extracted from the activation patterns of multiple involved muscles, has been applied successfully in detecting hand motion patterns based on multi-channel EMG signals [17], [18]. The recognition results can be used to control a variety of devices [19]–[21] including prostheses and exoskeletons. Myoelectric pattern recognition has significantly improved the dexterity of prosthesis control for amputee patients [22], particularly for those patients who had a surgery called targeted muscle reinnervation [23], [24]. It is therefore one of the advances to provide dexterous control of multi-DOF movements (e.g., an exoskeleton robot or a functional electrical stimulation system). A recent pilot study applied myoelectric pattern recognition to train a stroke subject's hand

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TABLE I
DETAILS OF THE EIGHT STROKE SUBJECTS

Subject	Age	Years since Onset	Gender	FMA [a]	FMA-C [a]	Grip (Kg) [b]		Pinch (Kg) [b]	
						Right	Left	Right	Left
S1	77	13	M	17	2	4.4	18.6	1.5	4.0
S2	53	4	M	29	0	2.2	36.4	0	8.5
S3	66	8	F	40	4	10.5	25.8	3.5	6.5
S4	69	7	M	21	0	11.2	42.2	3.5	9.5
S5	63	4	F	39	7	6.3	24.3	3.5	6.0
S6	41	1	M	38	1	7.6	30.3	5.5	14.5
S7	44	5	F	57	7	15.8	34.4	3.5	9.0
S8	50	1	M	16	0	1.8	25.1	1.0	6.0

[a]FMA = Fugl-Meyer Assessment Upper Extremity, a score between 0 (no function) to 66 (intact); FMA-C = Part C (Hand) of FMA, a score between 0 (no function) to 14 (intact).
[b]Grip force was measured using Jamar Plus+ Digital Hand Dynamometer (Patterson Medical, Warrenville, IL) with 0.1 Kg resolution; pinch force was measured using PG-60 Pinch Gauges (B&L Engineering, Santa Ana, CA) with 0.5 Kg resolution.

function using an exoskeleton driven by the subject's own motion intents, and observed dramatic hand function improvements [25]. However, the feasibility of applying myoelectric pattern recognition-based control on stroke patients has not been systematically investigated and thus still remains unclear. There are a few studies reporting stroke subjects' motion recognition performance based on offline myoelectric pattern recognition, but showing very different results. For example, Geng *et al.* [26] reported high accuracies (above 79%) from all three stroke subjects with hemiparesis, and the average accuracy was 5 times as high as the chance level. Ramos-Murguialday *et al.* [27] discovered that only one fifth of the subjects achieved an acceptable classification accuracy (above 65%) of three hand motion patterns, and the average accuracy was only 2 times as high as the chance level. Cesqui *et al.* [28] reported large individual differences in accuracies (from 36.7% to 83.3%), and the average accuracy was slightly above 2 times as high as the chance level. Lee *et al.* [29] reported mean classification accuracy of 71.3% for moderately impaired stroke subjects and 37.9% for severely impaired stroke subjects for classification of six hand tasks. Zhang *et al.* [30] applied high-density surface EMG recording, and obtained a very high average accuracy of 96.1% from 12 hemiparetic stroke subjects. Few studies on real-time pattern recognition-driven control have been reported. A stroke subject who achieved an accuracy of 75% (for controlling a robotic hand, 4.5 times as high as the chance level) was reported in our previous study [25]. Considering that there is usually a gap between online and offline recognition accuracies (even for neurologically intact subjects) [31], [32], and two algorithms with similar offline accuracies can perform differently online [33], the feasibility of applying myoelectric pattern recognition control in stroke patients deserves further investigation.

In order to provide stroke patients with the benefit of advanced myoelectric control, this study investigates the performance of myoelectric pattern recognition on patients after a severe to a mild stroke. An EMG-driven exoskeleton hand designed for stroke rehabilitation [8] is controlled in real time by the muscle activation patterns on the patient's affected hand, providing real-time motion assistance. Both the offline recognition accuracy and real-time control accuracy are investigated to assess the feasibility of applying myoelectric pattern recognition-based control on stroke patients.

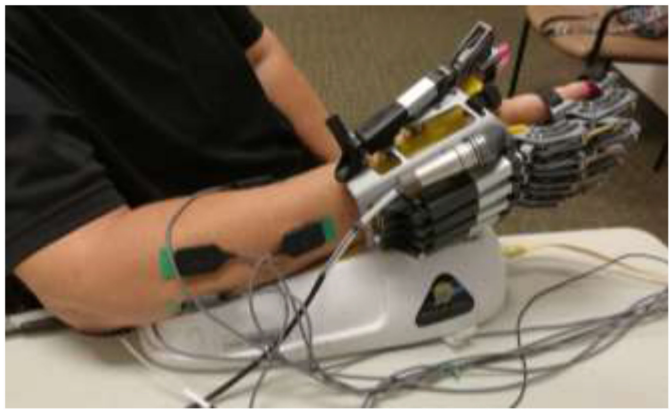

Fig. 1. A snapshot of the exoskeleton hand and the experiment setup.

II. METHODS

A. Subjects

Eight post-stroke subjects (S1-S8, 57.9 ± 12.0 years, 5 male, 3 female) participated in this study. All the subjects were right-handed before stroke and their right hand was affected after stroke. We chose right side affected subjects because the exoskeleton hand used in this study is designed for the right hand. All subjects were able to follow all the instructions and to give informed consent. These subjects covered a large range of impairment severity from a severe stroke with a score of 16 out of 66 in Fugl-Meyer Assessment Upper Extremity (FMA) to a mild stroke with an FMA score of 57. Details of these subjects are listed in Table I. All subjects gave written informed consent before any experiment procedures. The experimental protocols were approved by the Committee for the Protection of Human Subjects (CPHS) of the University of Texas Health Science Center at Houston and TIRR Memorial Hermann Hospital (Houston, TX, USA).

B. Exoskeleton Hand

An exoskeleton hand, Hand of Hope (Rehab-Robotics, Hong Kong), designed for stroke rehabilitation [8], [9], was employed in this study to assess the performance of real-time control driven by myoelectric pattern recognition. As shown in Fig. 1, the exoskeleton hand has a palm and five individual digits. The

subject's palm and five digits are fixed to the exoskeleton hand with Velcro belts, while the subject's forearm is supported by a brace. The subject therefore needs no effort to resist gravity, and the hand can be completely relaxed in the exoskeleton. The exoskeleton hand is also locked on the brace, restricting the subject's wrist and forearm movements (e.g., wrist flexion/extension and forearm rotation), so that the subject can maintain the same hand position during the experiment. Each digit of the exoskeleton hand is driven by a linear actuator. Linear movements are then converted into smooth rotations of the virtual metacarpophalangeal (MCP) joint and the virtual proximal interphalangeal (PIP) joint by the mechanical design to perform a natural finger/thumb flexion or extension. The exoskeleton can be controlled by commands from a computer through USB connection, and can be commanded to either maintain its position when the subject performs isometric contractions, or help the subject accomplish desired hand movements if the subject cannot perform the movement or if the movement range is limited.

C. Protocols

1) EMG Data Acquisition: In each experiment, the subject was seated comfortably on a chair with his/her right hand attached to the exoskeleton hand and locked in the neutral position. The exoskeleton hand was placed on a table next to the subject. The height of the table was adjusted to have the angle between the subject's upper arm and his trunk approximately 45 degrees. The subject's skin was first cleaned using sterile alcohol wipes. Seven bipolar surface electrodes (Delsys 2.1, Delsys Inc, Boston, USA) were then attached on the subject's forearm using double-sided tapes. Considering stroke induced muscle weakness, spasticity, atrophy, and other associated muscle internal structural and mechanical property changes, the exact position of the electrodes was personalized for each of the stroke subjects, targeting to record EMG signals from the first dorsal interosseous (FDI), flexor digitorum superficialis (FDS), flexor digitorum profundus (FDP), extensor digitorum (ED), abductor pollicis longus (APL), extensor digiti minimi (EDM), and extensor pollicis longus (EPL) muscles. In order to determine appropriate electrode locations, for each stroke subject an electrode was first placed on the skin of the targeted muscle. For the deep muscles, the skin above the muscle was determined according to muscle structure. For example, the FDP is approximately 1/4 distal to olecranon on the ulna side, and the APL is approximately 1/3 proximal to wrist on the radius side. Surface EMG signals were recorded during voluntary activation of the targeted muscle. Then, the electrode was detached from the skin and moved to several different nearby positions to record surface EMG signals of the same muscle. The final position of the electrode was visually determined as the one where the highest signal to noise ratio (SNR) was obtained. The reference electrode was placed on the olecranon, away from active muscles. The raw EMG signals were filtered by a 20–450 Hz band pass filter and amplified 10000 times by a Bagnoli-8 EMG System (Delsys Inc., Boston, USA). The signals were then digitized at a sampling rate of 1000 Hz with a 16-bit resolution by a data acquisition device USB-6221 (National Instruments Inc., Austin, USA). An application program developed using Microsoft Visual Studio was deployed on a desktop running Windows 7 to acquire and manage the EMG data. The exoskeleton hand was also connected to this desktop.

Each experiment consisted of two sessions as shown in Fig. 2. In the first session, EMG data were recorded for offline myoelectric pattern analysis. In the second session, EMG data were classified online and used to control the exoskeleton hand in real time. Six motion patterns of the exoskeleton hand were used in this study, including hand closing (HC); hand opening (HO); thumb, index and middle fingers closing (TIMC, or tripod pinch); thumb, index and middle fingers opening (TIMO); middle, ring and little fingers closing (MRLC, or the "gun" sign); and middle, ring and little fingers opening (MRLO). Each subject was given time to familiarize himself/herself with these motion patterns before the first session. Each subject also had sufficient rest between the two sessions to avoid mental and muscle fatigue.

2) Session 1: EMG Recording for Offline Analysis: In the first session, the subject was instructed to perform each motion pattern using a comfortable force level, preferably around 30%–40% of maximum voluntary contraction for a total of 15 times, each time lasting for about 2 seconds, while EMG data were recorded as data set DS-1. In order to cover more intra-pattern variations, the subject was not allowed to perform the same motion pattern two or more times in a row. Instead, the subject performed each motion pattern once in a random order (6 movements totally), and repeated 15 times. A short break was given after each movement. The position of the exoskeleton was locked during performance of each movement, so the EMG signals were recorded from isometric contractions. Before performing any closing movement (i.e., HC, TIMC, and MRLC), the experimenter set the exoskeleton hand to a fully open position, and before performing any opening movement (i.e., HO, TIMO, and MRLO), the experimenter set the exoskeleton hand to a fully closed position. The exoskeleton hand was then commanded to maintain its position (either fully open or fully closed) when EMG was being recorded. As a result, EMG signals of the 15 repetitions of each motion pattern were recorded under the same conditions (position of the subject's hand, finger, forearm, and also the electrodes), even though they were recorded separately.

3) Session 2: Online Motion Recognition and Control: In the second session, the subject performed each motion pattern using comfortable force for a total of 10 times in a random order, while EMG data were recorded as data set DS-2. The experimenter set the exoskeleton hand to a fully open position before the subject performed any closing movement, and to a fully closed position before the subject performed any opening movement. Therefore, the position of performing each movement in the second session was the same as that in the first session. Different from the first session, a classifier trained by DS-1 (the subject's own EMG data from the first session), was now employed to classify each movement. If a movement was detected and classified, the exoskeleton hand was then commanded to perform the recognized motion in real time. The subject was advised to keep performing a movement using a stable force until the exoskeleton hand began to move, and then relax and follow the exoskeleton hand passively. Only EMG

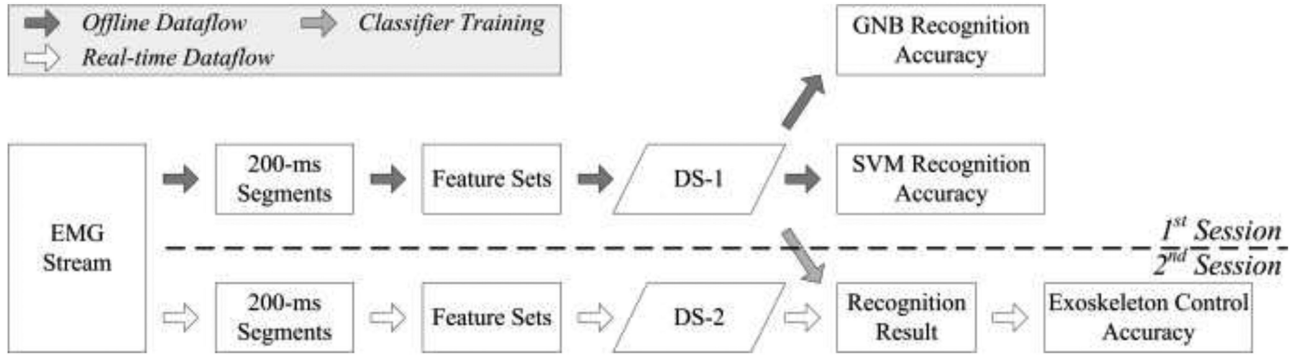

Fig. 2. Data flow diagram of the offline and real-time analysis.

data acquired during isometric contraction (i.e., before the exoskeleton hand moved) were used to control the exoskeleton hand. Once the exoskeleton hand began to move, it went all the way to the full range of motion. Consequently, the subject performed all the movements in a "normalized" hand position, either fully open or fully closed.

D. Myoelectric Pattern Recognition Algorithms

In both offline and online processing, EMG data were segmented into 200 ms analysis windows with 100 ms overlapping. Processing procedures including motion detection, feature extraction, and classification were performed on each of the analysis windows. For example, 9 analysis windows could be obtained from a 1-second movement, each providing a feature vector that would be used either as training data (to train the classifier) or as testing data (to be classified by a trained classifier). Such a window-based segmentation [34]–[36] can generate more data to train the classifier than a movement-based segmentation [37], [38], which generates only one sample per movement. When applied in real-time control, the updating rate of pattern recognition results or control commands of window-based segmentation is typically 10 Hz [34]–[36] (the same rate in this study). In contrast, the updating rate of movement-based segmentation is determined by the number of movements performed within one second [37], [38], usually below 1 Hz.

For each analysis window, motion detection was first performed to determine whether it was a valid data segment or just a baseline. An analysis window was considered to contain subject's movement information if the average mean absolute value [35] (MAV) of all the seven channels was greater than a given threshold (which was 80% of the average MAV of EMG signals recorded from the same subject at his/her comfortable force level), and vice versa. A set of features including root mean square (RMS) amplitude [35], 4th order auto regressive (AR) coefficients [20], [39], and waveform length [40] (WL) were then calculated based on each active or valid analysis window to form a feature vector. Support vector machines [41] (SVM) and Gaussian Naive Bayes classifier [35], [37] (GNB) with equal-prior-probability assumption were applied to classify motion patterns based on each feature vector, respectively.

E. Offline Motion Pattern Discrimination Assessment

The offline myoelectric pattern recognition performance was assessed using DS-1 by calculating the classification accuracy. Two strategies, namely the random cross validation (RCV) and chronological handout validation (CHV), were used to form the training and testing data sets. For RCV, 80% of the data were randomly picked out from each motion pattern of a subject to form the training data sets, while the subject's remaining 20% formed the testing data sets. More specifically, data of each pattern from a single subject were randomly divided into 10 groups, k-fold cross validation was then applied. The average accuracy of each classifier was estimated by repeating such calculations 50 times on each subject using randomly grouped data each time. For CHV, the first 80% (in chronological order) of the data from each motion pattern of a subject formed the training data sets, while the subject's remaining 20% formed the testing data sets. The analysis was implemented in Matlab 2014 (The MathWorks Inc., Natick, USA) with LIBSVM [41] for Matlab.

F. Online Motion Recognition and Exoskeleton Control

In the online motion recognition and control, a SVM classifier (implemented based on LIBSVM for C++) trained using DS-1 was applied to recognize the subject's movements (DS-2) with aforementioned EMG features. A program was developed using Microsoft Visual Studio to manage the process in a real-time manner. The program fills analysis windows with EMG data stream, and triggers the motion detection algorithm when an analysis window is filled. The program then launches the feature extraction and the SVM classifier if a movement is detected in this analysis window, and requests the exoskeleton hand to perform the same movement as the recognition result (i.e., sends a movement request) before another analysis window is filled. In order to increase the control accuracy, a two-window voting strategy was applied when controlling the exoskeleton hand: the latest movement request is performed only when it matches the movement request generated from the previous analysis window. Inexecutable movement requests (e.g., HC when the exoskeleton hand is closed) were discarded. There is one analysis window filled every 100 ms (or 10 Hz as mentioned in Section II.D), while our program only needs about 50 ms (on an

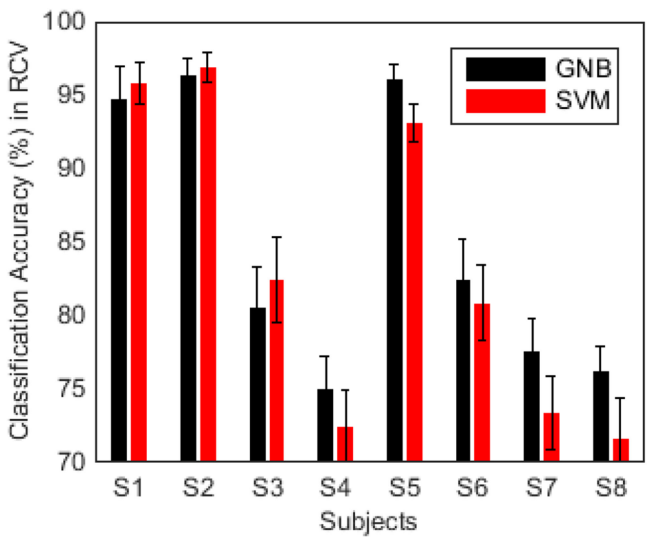

Fig. 3. Classification accuracy in random cross validation.

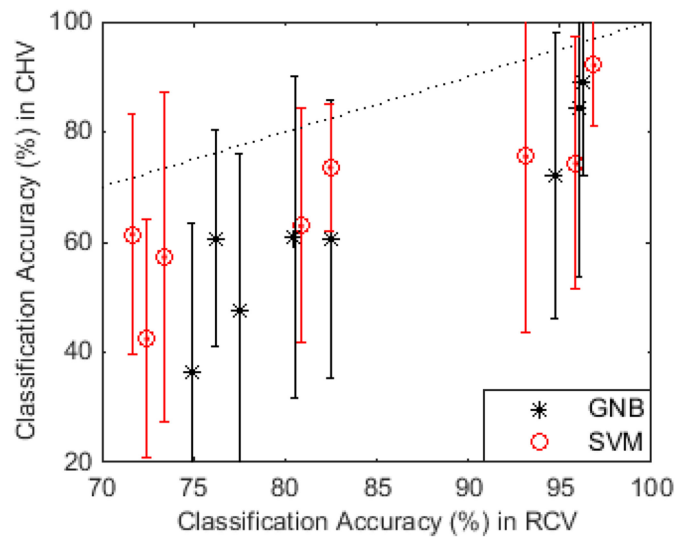

Fig. 4. Recognition accuracy in chronological handout validation. The x-value of each black "$*$" is the classification accuracy achieved by one subject in RCV using GNB, and the y-value is the classification accuracy achieved by the same subject in CHV using GNB. The x-value of each red "o" is the classification accuracy achieved by one subject in RCV using SVM, and the y-value is the classification accuracy achieved by the same subject in CHV using SVM. The error bars are the standard deviation of the six motion patterns. The classification accuracy obtained from CHV is lower than that from RCV if the point is below the dotted line.

Intel Celeron 1007U CPU at 1.5 GHz) to obtain a recognition result. Therefore, every analysis window can be processed in time. During the experiment, the movements performed by the subject before the exoskeleton hand moved (isometric contraction) and those performed by the exoskeleton hand (also performed passively by the subject) were recorded, respectively. The control accuracy was calculated as the percentage of the consistent movements to the total movements.

III. Results

A. Offline Myoelectric Pattern Recognition Performance

Fig. 3 shows the average classification accuracy of each stroke subject obtained from RCV, and the error bars show the standard deviation of the 50 tests. Hand motion patterns from each subject were well classified with an accuracy above 70%. Similar accuracies were achieved when using different classifiers for each subject (the average classification accuracy of all the subjects were 84.8 $\pm$ 9.3% for GNB, 83.3 $\pm$ 10.7% for SVM, $p = 0.125$ in paired T-Test). Clear individual differences in classification performance were also observed. For example, three subjects achieved classification accuracies around 95%, whereas another three subjects achieved accuracies only about 75%.

Fig. 4 shows the average classification accuracies obtained from CHV using GNB and SVM, respectively, and the error bars show the standard deviation of the six motion patterns. Similar to observed in RCV, there is no significant difference in classification performance between the two classifiers (the average classification accuracy of all the subjects were 63.9 $\pm$ 17.6% for GNB, 67.4 $\pm$ 14.8% for SVM, $p = 0.164$ in paired T-Test). However, all the subjects achieved a significantly lower classification accuracy in CHV than in RCV ($p < 0.001$ for both GNB and SVM in paired T-Test). And only three and four stroke subjects achieved a classification accuracy above 70% using GNB and SVM, respectively.

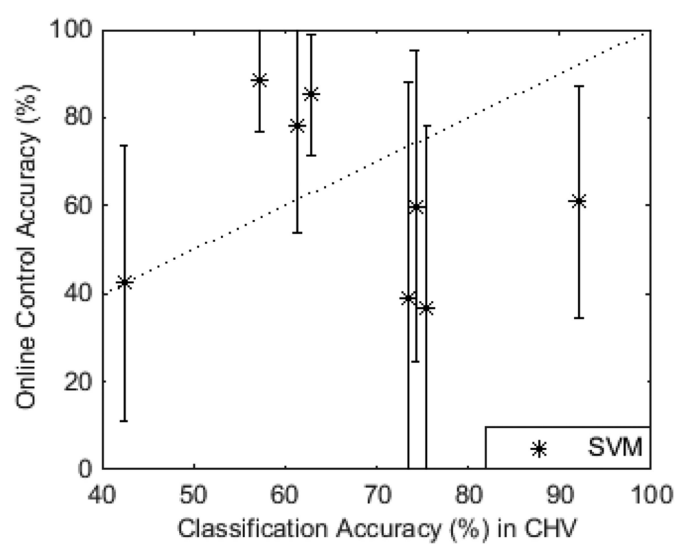

Fig. 5. Accuracy of online exoskeleton hand control. The x-value of each black "$*$" is the classification accuracy achieved by one subject from CHV using SVM, and the y-value is the control accuracy achieved by the same subject in the online exoskeleton control using SVM. The error bars are the standard deviation of the six motion patterns. The control accuracy is lower than the classification accuracy obtained from CHV if the point is below the dotted line.

B. Accuracies in Online Myoelectric Pattern Recognition and Exoskeleton Control

The myoelectric pattern recognition control accuracy of each stroke subject for online controlling the exoskeleton hand is shown in Fig. 5. Three subjects achieved a control accuracy

above 70%, and the average control accuracy of all the subjects was $61.3 \pm 20.9\%$. When compared with the classification accuracies obtained from CHV, three subjects achieved a higher accuracy, four subjects achieved a lower accuracy, and one subject achieved almost the same accuracy (less than 0.1% difference) in controlling the exoskeleton hand. No significant difference was observed between the control accuracy and the classification accuracy obtained from CHV ($p = 0.547$ in paired T-Test). No clear linear correlation was observed between the control accuracy and the classification accuracy obtained from RCV (with Pearson correlation coefficient $r = -0.20$), the classification accuracy obtained from CHV ($r = -0.17$), FMA ($r = 0.21$), FMA-C ($r = -0.12$), right hand grip force ($r = 0.04$), or pinch force ($r = 0.08$). The three subjects with the lowest control accuracies failed to control the exoskeleton hand to perform one or more motion patterns, i.e., each of them achieved a control accuracy of 0% for one or more hand motion patterns. Two subjects achieved a control accuracy of 0% for MRLC, making it the most "difficult" hand motion. HC was the only hand motion that every subject performed with a non-zero control accuracy.

IV. Discussion

Lag and accuracy are the two major performance measurements of a real-time myoelectric control system. With the improvement of the computational power of modern processors, most of the popular pattern recognition algorithms can be applied in real-time control [25], [42], [43]. Nowadays mobile devices also have the power to perform complex EMG signal processing in real time [44], [45]. Therefore, implementing a myoelectric control algorithm within an acceptable lag is no longer a big challenge.

In this study, we focused on accuracy when assessing performance of a myoelectric pattern recognition control system in stroke. Previous studies reported quite different results about myoelectric pattern recognition accuracies [26]–[30], extrapolating conflicting views about the feasibility of applying such an advanced control strategy in stroke. Considering that the variations may arise from a variety of factors (such as different algorithms, different protocols, and different severities of stroke, etc.), we used the following designs in our study. Firstly, we applied two fundamentally different classifiers, namely the GNB and the SVM, the former working in the feature space based on estimation of class distributions, while the latter working in a higher dimensional space based mainly on the structure of samples. Secondly, we well controlled our experiment setup and protocols. For example, EMG signals of the motion HC may change with the finger angles [46], [47]. In our experiment, EMG signals of hand motion patterns of every subject were acquired in the same position (e.g., the joint angle and the muscle length) to avoid position-related intra-pattern variations. The subject's hand was locked in the exoskeleton (which is closer to practical application), so that there was little shift between electrodes and muscles. The subject's hand was also supported by the brace, so that there was no interference caused by anti-gravity movement. Considering that the EMG signals tended to be very similar when the subject repeatedly perform the same motion at a time, causing underestimation of intra-pattern variations, we asked the subject to perform a different motion pattern from the previous one. Thirdly, in order to investigate the feasibility of applying myoelectric pattern recognition in stroke population, we recruited stroke subjects covering a wide range of time after stroke onset as well as impairment severity with different residual hand functions (since these parameters might have an effect on EMG patterns [48]).

In a previous study, we used four channels of surface EMG of neurologically intact subjects to control the six patterns of the robotic hand [35]. Three more channels of EMG were used in the current study considering the challenge imposed by stroke induced neuromuscular changes. In contrast to the relatively fixed electrode positions in our previous study [35], the positions of the seven surface electrodes were not exactly the same for the tested stroke subjects, given that they were individually determined for each stroke subject based on examination of voluntary surface EMG signals. Although we adjusted electrode positions attempting to record from the targeted muscles, it is possible that some EMG channels might record significant signals from the neighboring muscles (i.e., cross-talk), especially for small or deep muscles such as APL. This would not be a major problem for myoelectric pattern recognition if the classification performance is not deteriorated. It has been reported that some interferences (such as heartbeat or involuntary motor activities) contaminating voluntary EMG will not necessarily decrease the myoelectric pattern recognition accuracy [49], [50]. This can be viewed as an advantage of the myoelectric pattern recognition control compared with the conventional myoelectric control (which is designed to map EMG signal from a single muscle to a single degree of freedom control).

Both random cross validation (RCV) and chronological hand-out validation (CHV) were used in this study to assess the myoelectric pattern recognition performance of six different hand patterns in stroke. RCV is widely used in offline myoelectric pattern recognition studies. CHV can be viewed as a special case of RCV when the randomly formed training data sets happen to be the first 80% of the data for each motion. It is worth noting that CHV is closer to the real-time control, where a classifier trained using only existing (or previous) data is applied to classify real-time (or future) data. Theoretically, no significant difference in performance between RCV and CHV is expected if the motion patterns are stable. However, practically this might not be the case. With unstable or time-variant motion patterns, differences in classification may be observed between RCV and CHV.

Consistent classification accuracies were obtained from RCV using both GNB and SVM classifiers on each stroke subject, indicating the robustness of the pattern recognition performance. All the subjects achieved a relatively high accuracy (>65% for six motion patterns), consistent with Geng *et al.*'s results [26]. Three subjects achieved a very high accuracy of approximately 95% (the same level as reported by Zhang *et al.* [30]). However, we found that classification accuracies obtained from CHV were significantly lower than those from RCV. Only half of the stroke subjects achieved a relatively high classification accuracy of approximately 65% from CHV. Consistent classification accuracies were still obtained from CHV using both GNB and SVM classifiers on each subject. The findings suggest that the motion patterns extracted from each subject was unstable. Although we

tried to minimize motion variations in our experiment design, for each hand motion pattern performed by a stroke subject there were still variations in the last 3 repetitions (used as testing data sets in CHV) compared with the first 12 repetitions (used as training data sets in CHV). Such variations resulted in significant decrease of classification accuracies obtained from CHV.

In the online exoskeleton hand control experiment, DS-1 (existing data) were used to train the classifier, and DS-2 (real-time data) were used to assess the performance of the myoelectric pattern recognition control. Although the subject had a break between the two experimental sessions, DS-1 and DS-2 were acquired under the same condition (e.g., hand position, electrode placement, and muscle lengths). In the online control session, the exoskeleton hand was controlled by the real-time myoelectric pattern recognition result, but the movements executed by the exoskeleton hand were not necessarily simply mapped from the pattern recognition results. On one hand, not all the movements are executable. For example, a fully closed hand can perform hand/finger/thumb opening movements but not hand/finger/thumb closing movements. On the other hand, the exoskeleton needs more than 3 seconds to move from a fully closed position to a fully open position, while 30 myoelectric pattern recognition results can be obtained during the 3 seconds (updated at 10 Hz). If those results are inconstant, mapping each recognition result to the exoskeleton would not assist the subject to accomplish a motion smoothly. Therefore, a real-time control strategy was designed and control accuracy was reported in this session as performance measurement, instead of pattern recognition or classification accuracy. Not surprisingly, no significant difference was observed between the real-time control accuracy and the offline classification accuracy obtained from CHV. Those four stroke subjects whose control accuracy was higher than (or equal to) their classification accuracy in CHV were believed to benefit from the two-window voting strategy, which discarded occasional classification errors and thus improved the control accuracy, suggesting that these subjects were able to maintain the desired motion pattern for at least 300 ms (two windows) to drive the exoskeleton. On the contrary, the other four subjects had highly time-variant patterns which could not be classified correctly by the classifier trained using history data, or could not maintain 300 ms for the desired motion pattern to trigger the exoskeleton hand.

The results, along with the previous studies [26]–[30], show that if voluntary EMG can be recorded from chronic stroke patient's affected muscles, myoelectric pattern recognition can be helpful in identification of movement intentions (with a classification accuracy significantly higher than chance level). In our experiments, relatively high accuracies were achieved from every subject in RCV through offline myoelectric pattern recognition analysis, including those with no hand function (scoring 0 in FMA-C) or with very weak contraction level (e.g., S2). However, there are large individual differences among the stroke subjects. For CHV which is more representative to practical application, the classification accuracy was significantly lower. As to real-time control of the exoskeleton hand, only a portion of the tested stroke patients (38% in this study) can achieve an acceptable performance (i.e., control accuracy) when using myoelectric pattern recognition. Unfortunately, no clear correlation was found between the real-time myoelectric pattern recognition control performance and any of the known factors including impairment severity, residual hand function, and any offline classification accuracy (not only RCV, but also CHV; even though there was no statistical difference between the CHV offline performance and the real-time online control performance). Stroke patients with relatively high real-time control accuracies are believed to be able to use a myoelectric pattern recognition-based control interface for active training, but they do not necessarily have relatively high offline classification accuracies or residual hand function. This indicates that stroke patients should be individually online tested for their feasibility of applying myoelectric pattern recognition. For those stroke patients with relatively low real-time control accuracies, their time-variant motion patterns are one of the major challenges to overcome for improving both the offline and online myoelectric pattern recognition performance.

The current version of the exoskeleton robotic hand used in this study can perform six motion patterns in three groups (HC/HO, TIMC/TIMO, and MRLC/MRLO), among which the first two groups are highly related to daily life activities. Although the third group also represents common motions in myoelectric pattern recognition or gesture-machine interface studies, it is more desirable to practice tasks in functional context (i.e., task-oriented training) for stroke patients.

V. Conclusion

In summary, this study investigated both offline and online myoelectric pattern recognition of six hand motion patterns using forearm EMG signals in paretic side of eight stroke subjects. High classification accuracies and individual differences were observed in the offline myoelectric pattern analysis using the random cross validation (RCV). The challenge from unstable motion patterns of stroke subjects was disclosed by the decreased classification accuracies obtained from the chronological handout validation (CHV). Furthermore, experiments on real-time myoelectric pattern recognition control of an exoskeleton hand were performed. The control accuracies indicate that it is feasible to apply such an advanced myoelectric control strategy in some but not all of the stroke patients for robot-assisted rehabilitation. Our study did not reveal a clear correlation between the real-time control accuracy and the offline myoelectric pattern recognition performance, or any specific characteristics of the stroke subjects. Therefore, we suggest each stroke subject should be individually online tested for assessing the feasibility of applying myoelectric pattern recognition control.

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